Sketch-and-Restart: Randomized Sketching in Quadrature-Based Restarting for Matrix Functions

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4 Numerical Experiments

What Is a Matrix Function?

Let $\boldsymbol{A} \in \mathbb{C}^{N\times N}$ be a given matrix, and let f be a scalar function with the following integral representation:

$$f(z) = \int_\Gamma \frac{g(t)}{t-z}\,\mathrm{d}t, \quad \Gamma \subset \mathbb{C}\setminus\Omega, \quad z \in \Omega \supset \Lambda(\boldsymbol{A}).$$

Examples:

- Cauchy integral formula: $f(z) = \frac{1}{2\pi i}\int_\Gamma \frac{f(t)}{t-z}\,\mathrm{d}t$.
- Stieltjes function: $f(z) = \int_{-\infty}^0 \frac{g(t)}{t-z}\,\mathrm{d}t$, where $g(t) \le 0$.
  - $z^{-\alpha} = \dfrac{\sin((\alpha-1)\pi)}{\pi}\displaystyle\int_{-\infty}^0 \frac{(-t)^{-\alpha}}{t-z}\,\mathrm{d}t$, $\alpha \in (0,1)$.
  - $\dfrac{\log(1+z)}{z} = \displaystyle\int_{-\infty}^{-1} \frac{t^{-1}}{t-z}\,\mathrm{d}t$.

What Is a Matrix Function?

Let $\boldsymbol{A} \in \mathbb{C}^{N\times N}$ be a given matrix, and let f be a scalar function with the following integral representation:

$$f(z) = \int_\Gamma \frac{g(t)}{t-z}\,\mathrm{d}t, \quad \Gamma \subset \mathbb{C}\setminus\Omega, \quad z \in \Omega \supset \Lambda(\boldsymbol{A}).$$

The matrix function $f(\boldsymbol{A})$ is defined as

$$f(\boldsymbol{A}) = \int_\Gamma g(t)(t\boldsymbol{I}-\boldsymbol{A})^{-1}\,\mathrm{d}t.$$

Goal: Compute $f(\boldsymbol{A})\boldsymbol{b}$ for a large and sparse matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b} \in \mathbb{C}^N$.

Polynomial Approximation and Krylov Subspaces

A polynomial approximation gives a vector in a small subspace:

$$f(\boldsymbol{A})\boldsymbol{b} \approx p_{m-1}(\boldsymbol{A})\boldsymbol{b} \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}),$$

where p_{m-1} is a polynomial of degree at most $m-1$ and

$$\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}) = \operatorname{span}\{\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \ldots, \boldsymbol{A}^{m-1}\boldsymbol{b}\}.$$

Main steps

- Construct a basis of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.
- Represent $p_{m-1}(\boldsymbol{A})\boldsymbol{b}$ in terms of the basis.

Classical Arnoldi Approximation

With $\beta = \|\boldsymbol{b}\|$ and $\boldsymbol{v}_1 = \boldsymbol{b}/\beta$, the Arnoldi process constructs

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m \boldsymbol{H}_m + \boldsymbol{v}_{m+1} h_{m+1,m} \boldsymbol{e}_m^\top,$$

where $\boldsymbol{V}_{m+1}$ is orthonormal and $\boldsymbol{H}_m$ is upper Hessenberg.
Then the Arnoldi (FOM) approximation (Higham 2008) of $f(\boldsymbol{A})\boldsymbol{b}$ is

$$\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m) \boldsymbol{V}_m^* \boldsymbol{b} = \underbrace{\boldsymbol{V}_m}_{N\times m} \underbrace{f(\boldsymbol{H}_m)}_{m\times m} \underbrace{\boldsymbol{e}_1 \beta}_{m\times 1}.$$

$\rightsquigarrow$ Only need to compute the small and dense matrix function $f(\boldsymbol{H}_m)$.

The Cost of a Long Krylov Basis

For a general nonsymmetric matrix, the j-th step of Arnoldi uses all previous vectors:

$$\begin{aligned}
&\widetilde{\boldsymbol{v}}_{j+1} = \boldsymbol{A}\boldsymbol{v}_j, \\
&\widetilde{\boldsymbol{v}}_{j+1} \leftarrow \widetilde{\boldsymbol{v}}_{j+1} - \boldsymbol{v}_i \langle \widetilde{\boldsymbol{v}}_{j+1}, \boldsymbol{v}_i \rangle, \quad i = 1, \ldots, j, \\
&\boldsymbol{v}_{j+1} = \widetilde{\boldsymbol{v}}_{j+1} / \|\widetilde{\boldsymbol{v}}_{j+1}\|.
\end{aligned}$$

Work over m steps	Cost	Possible remedy
Store the basis	$\mathcal{O}(Nm)$	Restarting
Orthogonalize the basis	$\mathcal{O}(Nm^2)$	Sketching / Truncation
Apply $\boldsymbol{A}$	m matrix–vector products	Depends on $\boldsymbol{A}$

Our aim: Combine inexpensive basis construction with repeated short cycles.

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Arnoldi Approximation in Linear Systems

For a linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, suppose we have the Arnoldi decomposition

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top, \qquad \boldsymbol{b} = \boldsymbol{v}_1\beta.$$

Arnoldi approximation:

$$\boldsymbol{x}_m = \boldsymbol{V}_m\boldsymbol{H}_m^{-1}\boldsymbol{e}_1\beta \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}).$$

The residual is

$$\boldsymbol{r}_m = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_m = -\boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top\boldsymbol{H}_m^{-1}\boldsymbol{e}_1\beta = \boldsymbol{v}_{m+1}\frac{\gamma_m\beta}{\det(-\boldsymbol{H}_m)},$$

where $\gamma_m = \prod_{j=1}^{m} h_{j+1,j}$.

Arnoldi Approximation in Linear Systems

Arnoldi approximation: $\boldsymbol{x}_m = \boldsymbol{V}_m \boldsymbol{H}_m^{-1} \boldsymbol{e}_1 \beta$.
Residual:

$$\boldsymbol{r}_m = -\boldsymbol{v}_{m+1} h_{m+1,m} \boldsymbol{e}_m^\top \boldsymbol{H}_m^{-1} \boldsymbol{e}_1 \beta = \boldsymbol{v}_{m+1} \frac{\gamma_m \beta}{\det(-\boldsymbol{H}_m)},$$

Cramer's rule:

$$\boldsymbol{e}_m^\top \boldsymbol{H}_m^{-1} \boldsymbol{e}_1 = \text{the } m\text{-th entry of } \boldsymbol{H}_m^{-1} \boldsymbol{e}_1 = \det \left(\begin{bmatrix} h_{1,1} & h_{1,2} & \cdots & h_{1,m-1} & 1 \\ h_{2,1} & h_{2,2} & \cdots & h_{2,m-1} & 0 \\ & h_{3,2} & \cdots & h_{3,m-1} & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ & & \cdots & h_{m,m-1} & 0 \end{bmatrix} \right) / \det(\boldsymbol{H}_m)$$

$$= (-1)^{m+1} \prod_{j=1}^{m-1} h_{j+1,j} / \det(\boldsymbol{H}_m).$$

Restarting in Linear Systems

The restarting process is

$$
\begin{aligned}
&\boldsymbol{x}_m \approx \boldsymbol{A}^{-1}\boldsymbol{b}, && \boldsymbol{x}_m \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}) \\
&\boldsymbol{r}_m = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_m, && \\
&\boldsymbol{z}_m \approx \boldsymbol{A}^{-1}\boldsymbol{r}_m, && \boldsymbol{z}_m \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{r}_m) = \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{v}_{m+1}), \\
&\boldsymbol{x}^+ = \boldsymbol{x}_m + \boldsymbol{z}_m. &&
\end{aligned}
$$

Restarting as Error Correction

Let $f(z) = z^{-1}$ and $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b} = f(\boldsymbol{A})\boldsymbol{b}$.
Note that

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{x}_m = \boldsymbol{A}^{-1}\boldsymbol{b} - \boldsymbol{x}_m = \boldsymbol{A}^{-1}\boldsymbol{r}_m = f(\boldsymbol{A})\boldsymbol{r}_m = f(\boldsymbol{A})\boldsymbol{v}_{m+1}\frac{\gamma_m\beta}{\det(-\boldsymbol{H}_m)}.$$

The restart step can therefore be written as

$$\begin{aligned}
&\boldsymbol{x}_m \approx f(\boldsymbol{A})\boldsymbol{b}, && \boldsymbol{x}_m \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}),\\
&\boldsymbol{z}_m \approx \operatorname{err}(\boldsymbol{A})\boldsymbol{v}_{m+1}, && \boldsymbol{z}_m \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{v}_{m+1}),\\
&\boldsymbol{x}^+ = \boldsymbol{x}_m + \boldsymbol{z}_m,
\end{aligned}$$

where

$$\operatorname{err}(z) = \frac{\gamma_m\beta}{\det(-\boldsymbol{H}_m)}f(z)$$

is the error function.

For matrix functions, the error function is not necessarily a scalar multiple of f.

An Alternative Expression of the Arnoldi Approximation

Recall the Arnoldi approximation of $f(\boldsymbol{A})\boldsymbol{b}$:

$$\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.$$

If $p_{m-1} \in \mathcal{P}_{m-1}$ interpolates f at $\Lambda(\boldsymbol{H}_m)$, then

$$\boldsymbol{f}_m = \boldsymbol{V}_m p_{m-1}(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.$$

Lemma (Paige, Parlett, and Vorst 1995)

Let $q_m(z) = c_0 + c_1 z + \cdots + c_m z^m \in \mathcal{P}_m$. *Then*

$$q_m(\boldsymbol{A})\boldsymbol{v}_1 = \boldsymbol{V}_m q_m(\boldsymbol{H}_m)\boldsymbol{e}_1 + \boldsymbol{v}_{m+1} c_m \gamma_m,$$

where $\gamma_m = \prod_{j=1}^{m} h_{j+1,j}$.

The proof of this lemma does not assume that $\boldsymbol{V}_{m+1}$ is orthonormal.

An Alternative Expression of the Arnoldi Approximation

Recall the Arnoldi approximation of $f(\boldsymbol{A})\boldsymbol{b}$:

$$\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.$$

If $p_{m-1} \in \mathcal{P}_{m-1}$ interpolates f at $\Lambda(\boldsymbol{H}_m)$, then

$$\boldsymbol{f}_m = \boldsymbol{V}_m p_{m-1}(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.$$

Lemma: $q_m(\boldsymbol{A})\boldsymbol{v}_1 = \boldsymbol{V}_m q_m(\boldsymbol{H}_m)\boldsymbol{e}_1 + \boldsymbol{v}_{m+1}c_m\gamma_m$.

$$\boldsymbol{f}_m = p_{m-1}(\boldsymbol{A})\boldsymbol{v}_1\beta = p_{m-1}(\boldsymbol{A})\boldsymbol{b}.$$

Hermite Interpolation Formula

Now

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = f(\boldsymbol{A})\boldsymbol{b} - p_{m-1}(\boldsymbol{A})\boldsymbol{b}.$$

Lemma (Frommer, Güttel, and Schweitzer 2014)

The polynomial p_{m-1} that interpolates $f(z) = \int_\Gamma g(t)/(t-z)\,\mathrm{d}t$ at $\{\theta_j\}_{j=1}^m$ is given by

$$p_{m-1}(z) = \int_\Gamma \left(1 - \frac{\phi_m(z)}{\phi_m(t)}\right) \frac{g(t)}{t-z}\,\mathrm{d}t,$$

where $\phi_m(z) = (z-\theta_1)\cdots(z-\theta_m)$.

Therefore,

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \left(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\right)\phi_m(\boldsymbol{A})\boldsymbol{b},$$

where $\phi_m(t) = \det(t\boldsymbol{I}_m - \boldsymbol{H}_m)$.

Restarting in Matrix Functions

Lemma: $q_m(\boldsymbol{A})\boldsymbol{v}_1 = \boldsymbol{V}_m q_m(\boldsymbol{H}_m)\boldsymbol{e}_1 + \boldsymbol{v}_{m+1} c_m \gamma_m$.

$$\phi_m(\boldsymbol{A})\boldsymbol{b} = \boldsymbol{V}_m \phi_m(\boldsymbol{H}_m)\boldsymbol{e}_1 \beta + \boldsymbol{v}_{m+1}\gamma_m \beta = \boldsymbol{v}_{m+1}\gamma_m \beta,$$

since $\phi_m(t) = \det(t\boldsymbol{I}_m - \boldsymbol{H}_m)$.
Together with the error formula, we have

$$\boxed{f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \gamma_m \beta \left(\int_\Gamma \frac{g(t)}{\phi_m(t)} (t\boldsymbol{I} - \boldsymbol{A})^{-1} \, \mathrm{d}t \right) \boldsymbol{v}_{m+1} = \mathrm{err}(\boldsymbol{A})\boldsymbol{v}_{m+1}.}$$

Restarting process

$$\begin{aligned} \boldsymbol{f}_m &\approx f(\boldsymbol{A})\boldsymbol{b}, \qquad & \boldsymbol{f}_m &\in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}), \\ \boldsymbol{g}_m &\approx \mathrm{err}(\boldsymbol{A})\boldsymbol{v}_{m+1}, \qquad & \boldsymbol{g}_m &\in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{v}_{m+1}), \\ \boldsymbol{f}^+ &= \boldsymbol{f}_m + \boldsymbol{g}_m. \end{aligned}$$

In practice, $\mathrm{err}(\boldsymbol{A})\boldsymbol{v}_{m+1}$ is approximated by quadrature.

Revisiting Krylov Subspace Methods

Step 1: Express the approximation of $f(\boldsymbol{A})\boldsymbol{b}$.
For f with an integral representation,

$$f(\boldsymbol{A})\boldsymbol{b} = \int_\Gamma g(t)(t\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{b}\,\mathrm{d}t = \int_\Gamma g(t)\boldsymbol{x}(t)\,\mathrm{d}t,$$

where $\boldsymbol{x}(t)$ solves the shifted linear system

$$(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}.$$

Approximate $\boldsymbol{x}(t)$ in the Krylov subspace:

$$\boldsymbol{x}(t) \approx \boldsymbol{x}_m(t) \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b}).$$

Then

$$\boldsymbol{f}_m = \int_\Gamma g(t)\boldsymbol{x}_m(t)\,\mathrm{d}t.$$

Step 1: Express the approximation of $f(\boldsymbol{A})\boldsymbol{b}$.

$$\boldsymbol{f}_m = \int_\Gamma g(t)\boldsymbol{x}_m(t)\,\mathrm{d}t,$$

where $\boldsymbol{x}_m(t) \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ approximately solves the shifted linear system $(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}$.

Step 2: Express the approximate solution of the shifted linear system.

Let

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top, \qquad \boldsymbol{b} = \boldsymbol{v}_1\beta.$$

Then the Arnoldi (FOM) approximation of $\boldsymbol{x}(t)$ is

$$\boldsymbol{x}_m(t) = \boldsymbol{V}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta.$$

Revisiting Krylov Subspace Methods

Step 1: Express the approximation of $f(\boldsymbol{A})\boldsymbol{b}$.

$$\boldsymbol{f}_m = \int_\Gamma g(t)\boldsymbol{x}_m(t)\,\mathrm{d}t,$$

where $\boldsymbol{x}_m(t) \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ approximately solves the shifted linear system $(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}$.

Step 2: Express the approximate solution of the shifted linear system.

$$\boldsymbol{x}_m(t) = \boldsymbol{V}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta.$$

Step 3: Combine the two steps together.

$$\begin{aligned}\boldsymbol{f}_m &= \int_\Gamma g(t)\boldsymbol{x}_m(t)\,\mathrm{d}t = \int_\Gamma g(t)\boldsymbol{V}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta\,\mathrm{d}t \\ &= \boldsymbol{V}_m \int_\Gamma g(t)(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta\,\mathrm{d}t = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.\end{aligned}$$

An Alternative Proof of the Correction Formula

For the Arnoldi approximation $\boldsymbol{f}_m$, the error can be expressed as

$$\begin{aligned} f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m &= \int_\Gamma g(t)(\boldsymbol{x}(t) - \boldsymbol{x}_m(t))\,\mathrm{d}t \\ &= \int_\Gamma g(t)(t\boldsymbol{I} - \boldsymbol{A})^{-1}\big(\boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}_m(t)\big)\,\mathrm{d}t \\ &= \int_\Gamma g(t)(t\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{r}_m(t)\,\mathrm{d}t. \end{aligned}$$

An Alternative Proof of the Correction Formula

The error of the Arnoldi approximation is

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \int_\Gamma g(t)(t\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{r}_m(t)\,\mathrm{d}t.$$

Recall: The residual of the Arnoldi approximation of the shifted linear system $(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x} = \boldsymbol{b}$ can be expressed as

$$\boldsymbol{r}_m(t) = \boldsymbol{v}_{m+1}\frac{\gamma_m\beta}{\det(t\boldsymbol{I}_m - \boldsymbol{H}_m)} = \boldsymbol{v}_{m+1}\frac{\gamma_m\beta}{\phi_m(t)}.$$

Together, we have

$$\boxed{f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \gamma_m\beta\left(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\right)\boldsymbol{v}_{m+1} = \mathrm{err}(\boldsymbol{A})\boldsymbol{v}_{m+1}.}$$

Randomized Sketching

A random sketching matrix $\boldsymbol{S}$ maps vectors in a subspace of a high-dimensional space $\mathcal{V} \subset \mathbb{C}^N$ to a lower-dimensional space $\mathbb{C}^s$, where $\dim(\mathcal{V}) \leq s \ll N$, while approximately preserving their norms:

$$(1-\varepsilon)\|\boldsymbol{v}\|^2 \leq \|\boldsymbol{S}\boldsymbol{v}\|^2 \leq (1+\varepsilon)\|\boldsymbol{v}\|^2, \quad \boldsymbol{v} \in \mathcal{V},$$

where $\varepsilon \in (0,1)$ is a distortion parameter.

We say two vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{V}$ are $\boldsymbol{S}$-orthogonal, denoted by $\boldsymbol{u} \perp_{\boldsymbol{S}} \boldsymbol{v}$, if $\langle \boldsymbol{S}\boldsymbol{u}, \boldsymbol{S}\boldsymbol{v} \rangle = 0$.

Sketched Arnoldi Approximation

For the Arnoldi approximation of the shifted linear system, the residual $\boldsymbol{r}_m(t) = \boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}_m(t)$ satisfies the Galerkin condition: $\boldsymbol{r}_m(t) \perp \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.
We replace it by the sketched Galerkin condition: $\boldsymbol{r}_m(t) \perp_{\boldsymbol{S}} \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.
Let $\boldsymbol{V}_m$ be a basis of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ with $\boldsymbol{b} = \boldsymbol{v}_1\beta$. Then it can be verified that

$$\boldsymbol{x}_m(t) = \boldsymbol{V}_m\big(t\boldsymbol{I}_m - (\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\big)^{-1}\boldsymbol{e}_1\beta$$

and

$$\boxed{\boldsymbol{f}_m = \boldsymbol{V}_m f\big((\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\big)\boldsymbol{e}_1\beta.}$$

The vector $\boldsymbol{f}_m$ is called the sketched Arnoldi (sArnoldi) approximation of $f(\boldsymbol{A})\boldsymbol{b}$ [Güttel and Schweitzer 2023].

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The Sketch-and-Restart Framework

1. **Build a basis.** Use an Arnoldi-type process to generate an Arnoldi-like decomposition.
2. **Compute a correction.** Evaluate the current scalar function on the small matrix.

$$\boldsymbol{g}^{[k]} \approx \boldsymbol{V}_m^{[k]} \mathrm{err}^{[k]}(\boldsymbol{H}_m^{[k]}) \boldsymbol{e}_1 \beta^{[k]}.$$

3. **Accumulate.** Set $\boldsymbol{f}^{[k]} = \boldsymbol{f}^{[k-1]} + \boldsymbol{g}^{[k]}$.
4. **Restart.** Update the error function, retain the final basis vector, and discard the old basis.

Initially, $\mathrm{err}^{[0]} = f$ and $\boldsymbol{f}^{[-1]} = \boldsymbol{0}$.

Sketched Arnoldi Process

Idea: Construct a basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ such that

$$\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}.$$

In the following process:

$$\boldsymbol{S}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \operatorname{span}\{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_j\}, \qquad \boldsymbol{P}_j = \boldsymbol{S}\boldsymbol{V}_j, \qquad \boldsymbol{P}_j^* \boldsymbol{P}_j = \boldsymbol{I}_j.$$

Initialize $\beta = \|\boldsymbol{S}\boldsymbol{b}\|$, $\boldsymbol{p}_1 = \boldsymbol{S}\boldsymbol{b}/\beta$ and $\boldsymbol{v}_1 = \boldsymbol{b}/\beta$.
At the j-th step, execute the middle row sequentially for $i = 1, \ldots, j$:

$$\begin{aligned} \widetilde{\boldsymbol{v}}_{j+1} &= \boldsymbol{A}\boldsymbol{v}_j, & \widetilde{\boldsymbol{p}}_{j+1} &= \boldsymbol{S}\widetilde{\boldsymbol{v}}_{j+1}, \\ \widetilde{\boldsymbol{p}}_{j+1} &\leftarrow \widetilde{\boldsymbol{p}}_{j+1} - \boldsymbol{p}_i \underbrace{\langle \widetilde{\boldsymbol{p}}_{j+1}, \boldsymbol{p}_i \rangle}_{h_{i,j}}, & \widetilde{\boldsymbol{v}}_{j+1} &\leftarrow \widetilde{\boldsymbol{v}}_{j+1} - \boldsymbol{v}_i h_{i,j}, \\ h_{j+1,j} &= \|\widetilde{\boldsymbol{p}}_{j+1}\|, \quad \boldsymbol{p}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1}/h_{j+1,j}, & \boldsymbol{v}_{j+1} &= \widetilde{\boldsymbol{v}}_{j+1}/h_{j+1,j}. \end{aligned}$$

Sketched Arnoldi Process Using Projections

Idea: Construct a basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ such that $\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}$.
For each j,

- $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}$;
- $\boldsymbol{S}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \operatorname{span}\{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_j\}$ with $\boldsymbol{P}_j = \boldsymbol{S}\boldsymbol{V}_j$ orthonormal;

$\boldsymbol{\Pi}_j = \boldsymbol{I} - \boldsymbol{V}_j \boldsymbol{P}_j^* \boldsymbol{S}$ is the oblique projector satisfying

$$\boldsymbol{\Pi}_j^2 = \boldsymbol{\Pi}_j, \qquad \boldsymbol{\Pi}_j \boldsymbol{V}_j = \boldsymbol{0}, \qquad \boldsymbol{S}\boldsymbol{\Pi}_j = \boldsymbol{S} - \boldsymbol{P}_j \boldsymbol{P}_j^* \boldsymbol{S} = (\boldsymbol{I} - \boldsymbol{P}_j \boldsymbol{P}_j^*)\boldsymbol{S}.$$

At the j-th step,

$$\begin{aligned} \boldsymbol{v}_{j+1} &= \boldsymbol{\Pi}_j(\boldsymbol{A}\boldsymbol{v}_j)c_{j+1}, \\ \boldsymbol{p}_{j+1} &= \boldsymbol{S}\boldsymbol{v}_{j+1} = \boldsymbol{S}\boldsymbol{\Pi}_j(\boldsymbol{A}\boldsymbol{v}_j)c_{j+1} = (\boldsymbol{I} - \boldsymbol{P}_j\boldsymbol{P}_j^*)(\boldsymbol{S}\boldsymbol{A}\boldsymbol{v}_j)c_{j+1} \perp \boldsymbol{P}_j. \end{aligned}$$

sArnoldi Approximation

The sArnoldi process generates an Arnoldi-like decomposition

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top, \qquad \boldsymbol{b} = \boldsymbol{v}_1\beta,$$

where $\boldsymbol{V}_{m+1}$ is $\boldsymbol{S}$-orthonormal and $\boldsymbol{H}_m$ is upper Hessenberg.
Recall that the sArnoldi approximation of $f(\boldsymbol{A})\boldsymbol{b}$ and its corresponding shifted linear system are

$$\boldsymbol{f}_m = \boldsymbol{V}_m f\big((\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\big)\boldsymbol{e}_1\beta, \qquad \boldsymbol{x}_m(t) = \boldsymbol{V}_m\big(t\boldsymbol{I}_m - (\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\big)^{-1}\boldsymbol{e}_1\beta.$$

Note that

$$(\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m) = (\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{S}\boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top) = \boldsymbol{H}_m.$$

Therefore, the sArnoldi approximation can be simplified as

$$\boxed{\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta,} \qquad \boldsymbol{x}_m(t) = \boldsymbol{V}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta.$$

Sketched Harmonic Arnoldi Process: Motivation

Consider the shifted linear system in the integral representation of $f(\boldsymbol{A})\boldsymbol{b}$:

$$(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}.$$

The approximate solution $\boldsymbol{x}_m(t) \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.
In Arnoldi approximation and sArnoldi approximation, if the basis $\boldsymbol{V}_{m+1}$ is generated by the Arnoldi process or the sArnoldi process, then

$$\boldsymbol{x}_m(t) = \boldsymbol{V}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta.$$

Question: Can we say something about the GMRES approximation?
Definition:

- harmonic Arnoldi (GMRES-like) approximation: $\boldsymbol{r}_m(t) \perp \boldsymbol{A}\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.
- sketched harmonic Arnoldi approximation: $\boldsymbol{r}_m(t) \perp_{\boldsymbol{S}} \boldsymbol{A}\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$.

Goal: Construct a basis such that the sHmArnoldi approximation has a simple expression.

Sketched Harmonic Arnoldi Process

Idea: Construct a basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ such that

$$\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\,\mathrm{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}.$$

In the following process:

$$\boldsymbol{S}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \mathrm{span}\{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_j\}, \qquad \boldsymbol{SA}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \mathrm{span}\{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_j\},$$
$$\langle \boldsymbol{z}_i, \boldsymbol{p}_j \rangle = 0, \quad i < j, \qquad \langle \boldsymbol{z}_j, \boldsymbol{p}_j \rangle = 1.$$

Initialize $\beta = \|\boldsymbol{Sb}\|$, $\boldsymbol{p}_1 = \boldsymbol{Sb}/\beta$ and $\boldsymbol{v}_1 = \boldsymbol{b}/\beta$.
At the j-th step, execute the middle row sequentially for $i = 1, \ldots, j$:

$$\widetilde{\boldsymbol{v}}_{j+1} = \boldsymbol{A}\boldsymbol{v}_j, \qquad \widetilde{\boldsymbol{p}}_{j+1} = \boldsymbol{S}\widetilde{\boldsymbol{v}}_{j+1}, \quad \boldsymbol{z}_j = \frac{\widetilde{\boldsymbol{p}}_{j+1}}{\langle \widetilde{\boldsymbol{p}}_{j+1}, \boldsymbol{p}_j \rangle},$$
$$\widetilde{\boldsymbol{p}}_{j+1} \leftarrow \widetilde{\boldsymbol{p}}_{j+1} - \boldsymbol{p}_i \underbrace{\langle \widetilde{\boldsymbol{p}}_{j+1}, \boldsymbol{z}_i \rangle}_{h_{i,j}}, \qquad \widetilde{\boldsymbol{v}}_{j+1} \leftarrow \widetilde{\boldsymbol{v}}_{j+1} - \boldsymbol{v}_i h_{i,j},$$
$$h_{j+1,j} = \|\widetilde{\boldsymbol{p}}_{j+1}\|, \quad \boldsymbol{p}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1}/h_{j+1,j}, \qquad \boldsymbol{v}_{j+1} = \widetilde{\boldsymbol{v}}_{j+1}/h_{j+1,j}.$$

Sketched Harmonic Arnoldi Process Using Projections

Idea: Construct a basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ of $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ such that $\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\,\mathrm{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}$.
For each j,

- $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \mathrm{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j\}$;
- $\boldsymbol{S}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \mathrm{span}\{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_j\}$ with $\boldsymbol{P}_j = \boldsymbol{S}\boldsymbol{V}_j$, and $\boldsymbol{S}\boldsymbol{A}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b}) = \mathrm{span}\{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_j\}$;
- $\boldsymbol{T}_j = \boldsymbol{Z}_j^* \boldsymbol{P}_j$ is lower triangular with diagonal entries 1;

$\boldsymbol{\Pi}_j = \boldsymbol{I} - \boldsymbol{V}_j \boldsymbol{T}_j^{-1} \boldsymbol{Z}_j^* \boldsymbol{S}$ is the oblique projector satisfying

$$\boldsymbol{\Pi}_j^2 = \boldsymbol{\Pi}_j, \qquad \boldsymbol{\Pi}_j \boldsymbol{V}_j = \boldsymbol{0}, \qquad \boldsymbol{S}\boldsymbol{\Pi}_j = \boldsymbol{S} - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Z}_j^* \boldsymbol{S} = (\boldsymbol{I} - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Z}_j^*)\boldsymbol{S}.$$

It can be verified that

$$(\boldsymbol{I} - \boldsymbol{p}_j \boldsymbol{z}_j^*) \cdots (\boldsymbol{I} - \boldsymbol{p}_1 \boldsymbol{z}_1^*) = (\boldsymbol{I} - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Z}_j^*),$$

At the j-th step,

$$\boldsymbol{v}_{j+1} = \boldsymbol{\Pi}_j (\boldsymbol{A}\boldsymbol{v}_j) c_{j+1}$$
$$\boldsymbol{p}_{j+1} = \boldsymbol{S}\boldsymbol{v}_{j+1} = \boldsymbol{S}\boldsymbol{\Pi}_j (\boldsymbol{A}\boldsymbol{v}_j) c_{j+1} = (\boldsymbol{I} - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Z}_j^*)(\boldsymbol{S}\boldsymbol{A}\boldsymbol{v}_j) c_{j+1} \perp \boldsymbol{Z}_j.$$

sHmArnoldi Approximation

The sHmArnoldi process generates an Arnoldi-like decomposition

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top, \qquad \boldsymbol{b} = \boldsymbol{v}_1\beta,$$

where $\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{V}_j$ for all $1 \le j \le m$, and $\boldsymbol{H}_m$ is upper Hessenberg.
Let $\boldsymbol{x}_m(t) \in \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ approximately solve the shifted linear system $(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}$.
The projection condition $\boldsymbol{r}_m(t) \perp_{\boldsymbol{S}} \boldsymbol{A}\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ leads to

$$\boldsymbol{x}_m(t) = \boldsymbol{V}_m\Big(t\boldsymbol{I}_m - \big((\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{V}_m)\big)^{-1}(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\Big)^{-1}\boldsymbol{e}_1\beta$$

and

$$\boldsymbol{f}_m = \boldsymbol{V}_m f\Big(\big((\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{V}_m)\big)^{-1}(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)\Big)\boldsymbol{e}_1\beta.$$

sHmArnoldi Approximation

Arnoldi-like decomposition: $\boldsymbol{AV}_m = \boldsymbol{V}_m \boldsymbol{H}_m + \boldsymbol{v}_{m+1} h_{m+1,m} \boldsymbol{e}_m^\top$ with $\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{AV}_j$.

sHmArnoldi approximation: $\boldsymbol{f}_m = \boldsymbol{V}_m f\Big(\big((\boldsymbol{SAV}_m)^*(\boldsymbol{SV}_m)\big)^{-1}(\boldsymbol{SAV}_m)^*(\boldsymbol{SAV}_m)\Big)\boldsymbol{e}_1\beta$

Note that

$$(\boldsymbol{SAV}_m)^*(\boldsymbol{SAV}_m) = (\boldsymbol{SAV}_m)^*(\boldsymbol{SV}_m\boldsymbol{H}_m + \boldsymbol{Sv}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top) = (\boldsymbol{SAV}_m)^*(\boldsymbol{SV}_m)\boldsymbol{H}_m.$$

Consequently,

$$\big((\boldsymbol{SAV}_m)^*(\boldsymbol{SV}_m)\big)^{-1}(\boldsymbol{SAV}_m)^*(\boldsymbol{SAV}_m) = \boldsymbol{H}_m.$$

Therefore, the sHmArnoldi approximation simplifies to

$$\boxed{\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.}$$

Adaptive Truncated Arnoldi Process

Idea: Orthogonalize against only the most recent t basis vectors. Use the sketch to decide when the short cycle should end.

Initialize $\beta = \|\boldsymbol{b}\|$, $\boldsymbol{v}_1 = \boldsymbol{b}/\beta$, and $\boldsymbol{P}_1 = \boldsymbol{S}\boldsymbol{v}_1$.

At step j, execute the middle row sequentially for $i = \max\{1, j-t+1\}, \ldots, j$:

$$\widetilde{\boldsymbol{v}}_{j+1} = \boldsymbol{A}\boldsymbol{v}_j,$$
$$\widetilde{\boldsymbol{v}}_{j+1} \leftarrow \widetilde{\boldsymbol{v}}_{j+1} - \boldsymbol{v}_i \underbrace{\langle \widetilde{\boldsymbol{v}}_{j+1}, \boldsymbol{v}_i \rangle}_{h_{i,j}},$$
$$h_{j+1,j} = \|\widetilde{\boldsymbol{v}}_{j+1}\|, \quad \boldsymbol{v}_{j+1} = \widetilde{\boldsymbol{v}}_{j+1}/h_{j+1,j}, \qquad \boldsymbol{P}_{j+1} = \begin{bmatrix} \boldsymbol{P}_j & \boldsymbol{S}\boldsymbol{v}_{j+1} \end{bmatrix}.$$

Stop if $\kappa(\boldsymbol{P}_{j+1}) > \tau$ or the maximum cycle length is reached.

Rank-1 Update

After a truncated cycle, the final basis vector may not satisfy the projection condition needed for restarting. Choose $\boldsymbol{c}_m$ so that $\boldsymbol{v}_{m+1} = \boldsymbol{V}_m \boldsymbol{c}_m + \widetilde{\boldsymbol{v}}_{m+1}\alpha_{m+1}$ where $\widetilde{\boldsymbol{v}}_{m+1}$ satisfies that condition.

Then

$$\begin{aligned}\boldsymbol{A}\boldsymbol{V}_m &= \boldsymbol{V}_m \boldsymbol{H}_m + \boldsymbol{v}_{m+1} h_{m+1,m} \boldsymbol{e}_m^\top \\ &= \boldsymbol{V}_m \big(\boldsymbol{H}_m + \boldsymbol{c}_m h_{m+1,m} \boldsymbol{e}_m^\top\big) + \widetilde{\boldsymbol{v}}_{m+1}\big(\alpha_{m+1} h_{m+1,m}\big)\boldsymbol{e}_m^\top \\ &= \boldsymbol{V}_m \widetilde{\boldsymbol{H}}_m + \widetilde{\boldsymbol{v}}_{m+1}\tilde{h}_{m+1,m}\boldsymbol{e}_m^\top.\end{aligned}$$

Approximation	Required condition	$\boldsymbol{c}_m$
Arnoldi	$\widetilde{\boldsymbol{v}}_{m+1} \perp \boldsymbol{V}_m$	$\boldsymbol{V}_m^\dagger \boldsymbol{v}_{m+1}$
sArnoldi	$\widetilde{\boldsymbol{v}}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{V}_m$	$(\boldsymbol{S}\boldsymbol{V}_m)^\dagger(\boldsymbol{S}\boldsymbol{v}_{m+1})$
HmArnoldi	$\widetilde{\boldsymbol{v}}_{m+1} \perp \boldsymbol{A}\boldsymbol{V}_m$	$\big((\boldsymbol{A}\boldsymbol{V}_m)^* \boldsymbol{V}_m\big)^{-1}(\boldsymbol{A}\boldsymbol{V}_m)^* \boldsymbol{v}_{m+1}$
sHmArnoldi	$\widetilde{\boldsymbol{v}}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{V}_m$	$\big((\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{V}_m)\big)^{-1}(\boldsymbol{S}\boldsymbol{A}\boldsymbol{V}_m)^*(\boldsymbol{S}\boldsymbol{v}_{m+1})$

Rank-1 Update

After a truncated cycle, the final basis vector may not satisfy the projection condition needed for restarting. Choose $\boldsymbol{c}_m$ so that $\boldsymbol{v}_{m+1} = \boldsymbol{V}_m \boldsymbol{c}_m + \widetilde{\boldsymbol{v}}_{m+1}\alpha_{m+1}$ where $\widetilde{\boldsymbol{v}}_{m+1}$ satisfies that condition.

Then

$$\begin{aligned}
\boldsymbol{A}\boldsymbol{V}_m &= \boldsymbol{V}_m \boldsymbol{H}_m + \boldsymbol{v}_{m+1} h_{m+1,m} \boldsymbol{e}_m^\top \\
&= \boldsymbol{V}_m \big(\boldsymbol{H}_m + \boldsymbol{c}_m h_{m+1,m} \boldsymbol{e}_m^\top\big) + \widetilde{\boldsymbol{v}}_{m+1} \big(\alpha_{m+1} h_{m+1,m}\big) \boldsymbol{e}_m^\top \\
&= \boldsymbol{V}_m \widetilde{\boldsymbol{H}}_m + \widetilde{\boldsymbol{v}}_{m+1} \tilde{h}_{m+1,m} \boldsymbol{e}_m^\top.
\end{aligned}$$

After the rank-1 update, the approximation of $f(\boldsymbol{A})\boldsymbol{b}$ is still in the form of

$$\boxed{\boldsymbol{f}_m = \boldsymbol{V}_m f(\widetilde{\boldsymbol{H}}_m) \boldsymbol{e}_1 \beta.}$$

Restarting the Arnoldi-like Approximations

Under suitable conditions, all approximations of $f(\boldsymbol{A})\boldsymbol{b}$ can be expressed in a unified form:

$$\boxed{\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta,}$$

accompanied by an Arnoldi-like decomposition

$$\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_m\boldsymbol{H}_m + \boldsymbol{v}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top \qquad \boldsymbol{b} = \boldsymbol{v}_1\beta.$$

The restarting argument does not require $\boldsymbol{V}_{m+1}$ to be orthonormal.
Therefore,

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \gamma_m\beta\left(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\right)\boldsymbol{v}_{m+1} =: \mathrm{err}(\boldsymbol{A})\boldsymbol{v}_{m+1},$$

where $\gamma_m = \prod_{j=1}^m h_{j+1,j}$ and $\phi_m(t) = \det(t\boldsymbol{I}_m - \boldsymbol{H}_m)$.

Convergence Result [Güttel, L., and Nyman 2026]

Assumptions:

- f is a Stieltjes function: $f(z) = \int_{-\infty}^{0} g(t)/(t-z)\,\mathrm{d}t$ for a nonpositive function g;
- $\boldsymbol{A}$ is positive real: $\Re(\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}) > 0$ for every nonzero $\boldsymbol{x} \in \mathbb{C}^N$;
- $\boldsymbol{S}$ satisfies the ε-subspace embedding property: $\big|\langle \boldsymbol{u}, \boldsymbol{v}\rangle - \langle \boldsymbol{S}\boldsymbol{u}, \boldsymbol{S}\boldsymbol{v}\rangle\big| \le \varepsilon \|\boldsymbol{u}\|\|\boldsymbol{v}\|$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{K}_{m+1}\big(\boldsymbol{A}, \boldsymbol{v}_1^{[\ell]}\big)$;
- The sketched harmonic Ritz values satisfy $\Lambda(\boldsymbol{H}_m^{[\ell]}) \subset \mathbb{C}_+$.

Then the restarted sHmArnoldi approximant $\boldsymbol{f}^{[k]}$ satisfies

$$\big\|f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{[k]}\big\|_{\boldsymbol{A}^*\boldsymbol{A}} \le C\|\boldsymbol{b}\|\big(C_\varepsilon \alpha_m\big)^{k+1},$$

- C depends only on f and $\boldsymbol{A}$;
- $C_\varepsilon = \sqrt{(1+\varepsilon)/(1-\varepsilon)}$;
- $\alpha_m = \eta^m$ where $\eta \in [0,1)$ depends only on $\boldsymbol{A}$.

Geometric convergence follows whenever $C_\varepsilon \alpha_m < 1$.

1 Introduction

2 Restarting and Randomized Sketching in Matrix Functions

3 The Sketch-and-Restart Framework

4 Numerical Experiments

Convection-Diffusion Example ($f(z) = z^{-1/2}$, A nonsymm., $N = 10,000$)

method	error	time (s)	n_{iter}	n_{matvec}	n_{vec}
Arn	1.5e-08	1.7e+00	3	450	150
sArn	3.1e-09	1.2e+00	4	600	150
sHmArn	3.9e-09	1.2e+00	4	600	150
adaArn-2	1.5e-08	1.3e+00	3	450	150
adaArn-1	4.4e-08	8.3e-01	5	420	90
adaArn-0	2.8e-08	4.1e-01	19	263	15
adaHmArn-2	4.5e-08	1.2e+00	3	450	150
adaHmArn-1	1.4e-08	7.3e-01	4	355	90
adaHmArn-0	3.3e-08	4.1e-01	18	251	15

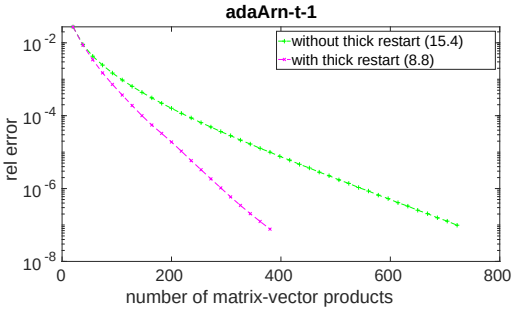
The $t = 0$ adaptive variants use the fewest stored vectors and have the shortest reported times.

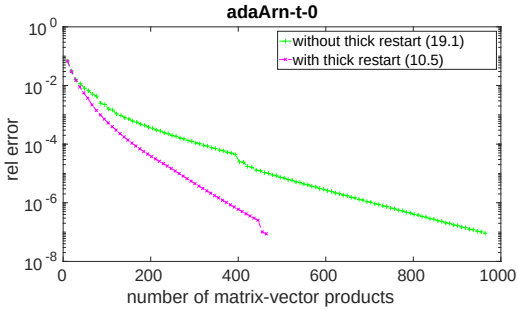
QCD Example ($f(z) = z^{-1/2}$, A nonsymm., $N = 49,152$)

method	error	time (s)	n_{iter}	n_{matvec}	n_{vec}
Arn	4.7e-10	1.4e+01	3	450	150
sArn	1.3e-09	1.3e+01	3	450	150
sHmArn	1.6e-09	1.3e+01	3	450	150
adaArn-2	7.8e-09	1.2e+01	4	410	105
adaArn-1	9.9e-08	1.5e+01	40	722	20
adaArn-0	9.1e-08	1.9e+01	106	964	10
adaHmArn-2	1.2e-08	1.2e+01	4	415	105
adaHmArn-1	9.6e-08	1.6e+01	41	740	20
adaHmArn-0	9.8e-08	2.8e+01	142	1279	10

$t = 2$ is the fastest truncated setting; $t = 0$ stores fewer vectors but requires much more time.

Thick Restarting Helps Aggressive Truncation in the QCD Example





For adaArn-0 and adaArn-1 in this QCD example, thick restarting reduces both the number of matrix–vector products and the reported runtime.

Take-Home Messages

- **A new sketched harmonic Arnoldi process.** It constructs an Arnoldi-like decomposition such that $\boldsymbol{v}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{V}_j$ for all $1 \leq j \leq m$.
- **One error-correction framework.** The Arnoldi-like decomposition and the appropriate final-vector condition give the same form $\boldsymbol{f}_m = \boldsymbol{V}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta$, which can be restarted.
- **A problem-dependent cost trade-off.** Adaptive truncation reduces full-vector orthogonalization, but an aggressive choice may require many more cycles.

Thank you!

My homepage:

`https://jingyuliumath.github.io/`



Related paper:

arXiv:2607.10354



Constants in the Convergence Result

Recall:

$$\left\| f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{[k]} \right\|_{\boldsymbol{A}^*\boldsymbol{A}} \leq C\|\boldsymbol{b}\| \left(C_\varepsilon \alpha_m \right)^{k+1},$$

Define three quantities

$$\begin{aligned}
\rho &= \min\left\{ \Re\left(\boldsymbol{v}^* \boldsymbol{A}^{-1} \boldsymbol{v} \right) : \boldsymbol{v} \in \mathbb{C}^N,\ \|\boldsymbol{v}\| = 1 \right\}, \\
\delta &= \min\left\{ |\boldsymbol{v}^* \boldsymbol{A} \boldsymbol{v}| : \boldsymbol{v} \in \mathbb{C}^N,\ \|\boldsymbol{v}\| = 1 \right\}, \\
\delta' &= \min\left\{ |\boldsymbol{v}^* \boldsymbol{A}^{-1} \boldsymbol{v}| : \boldsymbol{v} \in \mathbb{C}^N,\ \|\boldsymbol{v}\| = 1 \right\}.
\end{aligned}$$

Then

- $C = \|\boldsymbol{A}\| f\left(\rho \|\boldsymbol{A}\|^2\right)$;
- $C_\varepsilon = \sqrt{(1+\varepsilon)/(1-\varepsilon)}$;
- $\alpha_m = \eta^m$ with $\eta = \sqrt{1 - \delta\delta'} \in [0, 1)$.